Individual Final Project Report

Course: DATS 6312.11- NLP

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Title: Sentiment Analysis of Financial News Headlines

1.Introduction

Financial markets respond rapidly to the tone and sentiment conveyed in news headlines. Positive or negative shifts in wording often influence investor expectations, short-term price movements, and longer-term market behavior. However, the sheer volume of financial news produced daily makes manual interpretation inefficient and inconsistent. As part of our group project, we explored how Natural Language Processing (NLP) techniques can automatically evaluate financial headlines and classify their sentiment as *positive*, *negative*, or *neutral*. These sentiment labels can then be linked to market patterns to support data-driven forecasting and investment decisions.

We jointly developed the project design, selected the dataset, and decided on the overall direction of the project. My individual contribution focused on all modeling components such as building a classical baseline model, designing the Transformer-based pipeline, fine-tuning a DistilBERT classifier, evaluating performance, visualizing embedding behavior, and analyzing model errors. This report documents the technical details and results of the modeling portion.

For the presentation, we used a sample dataset of 10,000 rows because the full dataset contains 1.8 million rows, and we were working in PyCharm with Google Cloud resources. However, after receiving your feedback during the presentation, I attempted to use the entire dataset to avoid any issues during data preparation. Despite spending many hours, the GCP–PyCharm environment repeatedly failed when processing the full dataset. To overcome this and ensure smooth experimentation, I switched to Google Colab Pro and utilized its A100 GPU, which allowed me to train and test our models using the complete 1.8 million-row dataset.

The objective of this project is to automatically classify financial news headlines into positive, negative, or neutral sentiment using NLP methods, and to examine how these sentiment patterns may relate to market behavior.

1.Data Set Description:

DataSet:https://www.kaggle.com/datasets/miguelaenlle/massive-stock-news-analysis-db-for-nlpbacktests?select=raw_partner_headlines.csv

```
Shape (rows, columns): (1845559, 6)

Columns:
['Unnamed: 0', 'headline', 'url', 'publisher', 'date', 'stock']

Data types and non-null counts:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1845559 entries, 0 to 1845558
Data columns (total 6 columns):
 #   Column      Dtype
---  ------      -----
 0   Unnamed: 0  int64
 1   headline    object
 2   url         object
 3   publisher   object
 4   date        object
 5   stock       object
dtypes: int64(1), object(5)
memory usage: 84.5+ MB
None
```

We use the "Massive Stock News Analysis Database for NLP Backtests" dataset from Kaggle, specifically the raw_partner_headlines.csv file as it is shown in Fig 1. This dataset contains approximately 1.8 million English-language news headlines related to publicly traded stocks. Each row corresponds to a single news headline and includes metadata such as the publication date, time, associated ticker symbol, the headline text, and the news provider/source.

This large-scale dataset is designed for financial NLP tasks and backtesting strategies, making it well suited for our goal of financial sentiment classification. In our project, we first experimented with a 10,000-row sample for faster prototyping and debugging. After incorporating presentation feedback, I moved to using the full 1.8M-row dataset to train and evaluate our models more reliably and to reduce sampling bias.

	Unnamed: 0	headline	url	publisher	date	stock
0	2	Agilent Technologies Announces Pricing of $5........	http://www.gurufocus.com/news/1153187/agilent-...	GuruFocus	2020-06-01 00:00:00	A
1	3	Agilent (A) Gears Up for Q2 Earnings: What's i...	http://www.zacks.com/stock/news/931205/agilent...	Zacks	2020-05-18 00:00:00	A
2	4	J.P. Morgan Asset Management Announces Liquida...	http://www.gurufocus.com/news/1138923/jp-morga...	GuruFocus	2020-05-15 00:00:00	A
3	5	Pershing Square Capital Management, L.P. Buys ...	http://www.gurufocus.com/news/1138704/pershing...	GuruFocus	2020-05-15 00:00:00	A
4	6	Agilent Awards Trilogy Sciences with a Golden ...	http://www.gurufocus.com/news/1134012/agilent-...	GuruFocus	2020-05-12 00:00:00	A

Figure 1.1 raw data set head

Our dataset was originally unlabeled, meaning it did not contain any sentiment annotations required for supervised sentiment analysis. To create high-quality labels suitable for training modern NLP models, I have used **FinBERT**, a transformer model specifically designed for financial sentiment understanding. FinBERT generates sentiment predictions (positive, neutral, negative) based on financial-language patterns, which allowed us to automatically label all 1.8 million headlines as it is shown in Figure 1.2.

Unnamed: 0	headline	url	publisher	date	stock	sentiment
2	Agilent Technologies Announces Pricing of $5........	http://www.gurufocus.com/news/1153187/agilent-...	GuruFocus	2020-06-01 00:00:00	A	neutral
3	Agilent (A) Gears Up for Q2 Earnings: What's i...	http://www.zacks.com/stock/news/931205/agilent...	Zacks	2020-05-18 00:00:00	A	neutral
4	J.P. Morgan Asset Management Announces Liquida...	http://www.gurufocus.com/news/1138923/jp-morga...	GuruFocus	2020-05-15 00:00:00	A	negative
5	Pershing Square Capital Management, L.P. Buys ...	http://www.gurufocus.com/news/1138704/pershing...	GuruFocus	2020-05-15 00:00:00	A	neutral
6	Agilent Awards Trilogy Sciences with a Golden ...	http://www.gurufocus.com/news/1134012/agilent-...	GuruFocus	2020-05-12 00:00:00	A	positive
7	Agilent Technologies Inc (A) CEO and President...	http://www.gurufocus.com/news/1132985/agilent-...	GuruFocus	2020-05-11 00:00:00	A	neutral
8	' Stocks Growing Their Earnings Fast	http://www.gurufocus.com/news/1129235/3-stocks...	GuruFocus	2020-05-07 00:00:00	A	positive
9	Cypress Asset Management Inc Buys Verizon Comm...	http://www.gurufocus.com/news/1128926/cypress-...	GuruFocus	2020-05-07 00:00:00	A	neutral
10	Hendley & Co Inc Buys American Electric Power ...	http://www.gurufocus.com/news/1126951/hendley-...	GuruFocus	2020-05-05 00:00:00	A	neutral

Figure 1.2. Labeled data head

To evaluate how effective these automatically generated labels were, I introduced a **benchmark dataset**—the *Financial PhraseBank* from Hugging Face (https://huggingface.co/datasets/takala/financial_phrasebank). This dataset is manually curated by financial experts and is a widely accepted standard for financial sentiment analysis. Using our evaluation pipeline, FinBERT achieved strong performance on PhraseBank, confirming that it is a reliable teacher model for large-scale labeling. We also compared the sentiment distribution of our automatically labeled 1.8M-row dataset with the distribution in Financial PhraseBank and observed similar overall trends, which further supports the consistency of the labeling process.

In addition, I have analyzed FinBERT's confidence scores, identified low-confidence or ambiguous headlines, and examined keyword patterns and temporal sentiment trends. Together, these checks confirmed that our pseudo-labels are stable enough for downstream model training. FinBERT achieves 97.2% accuracy on the expert-annotated Financial PhraseBank dataset, showing excellent agreement with human labels. This validates FinBERT as a reliable model for generating high-quality sentiment labels for our 1.8M-row dataset.

```
Accuracy: 0.9717314487632509

Classification report:
              precision    recall  f1-score   support

    negative      0.906     0.983     0.943       303
     neutral      0.999     0.967     0.982      1391
    positive      0.947     0.977     0.962       570

    accuracy                          0.972      2264
   macro avg      0.951     0.976     0.963      2264
weighted avg      0.973     0.972     0.972      2264


Confusion matrix (rows=true, cols=pred):
[[ 298    1    4]
 [  19 1345   27]
 [  12    1  557]]
```

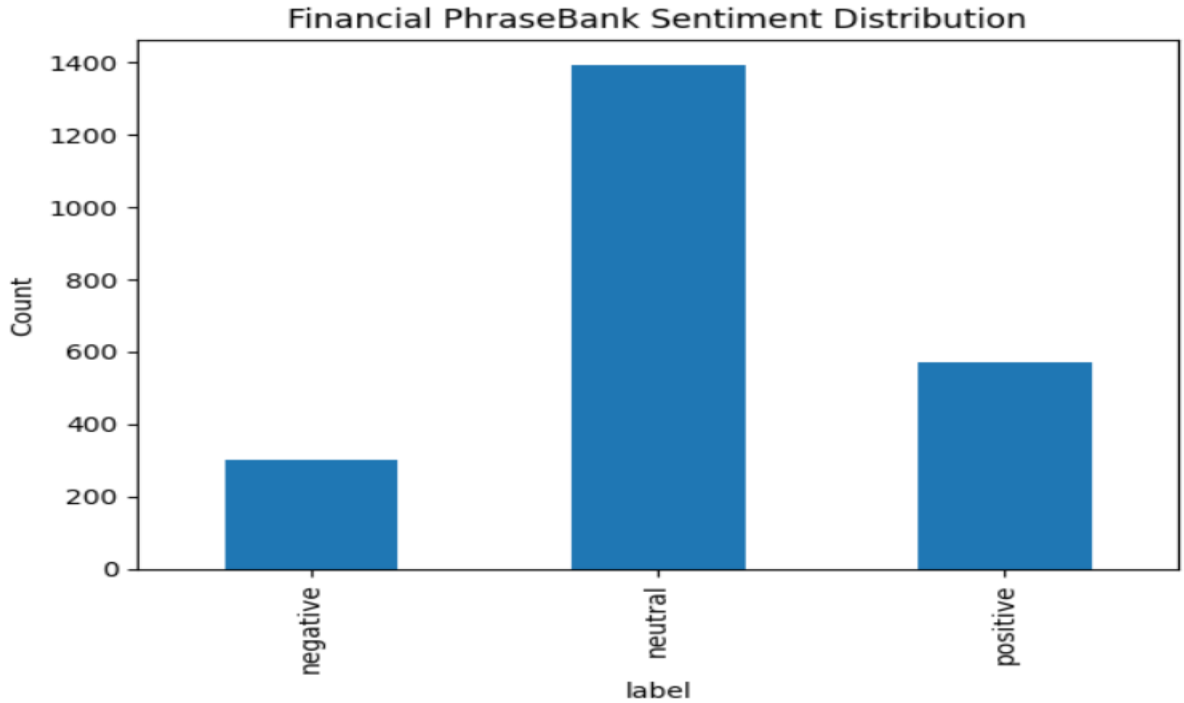


Figure 1.3. PhraseBank data set Sentiment distribution

The Financial PhraseBank dataset shows a similar pattern with neutral sentences being the largest category, followed by positive and negative statements. This similarity in sentiment proportions suggests that FinBERT's labeling behavior on the large dataset aligns well with the distribution found in a human-annotated financial benchmark.

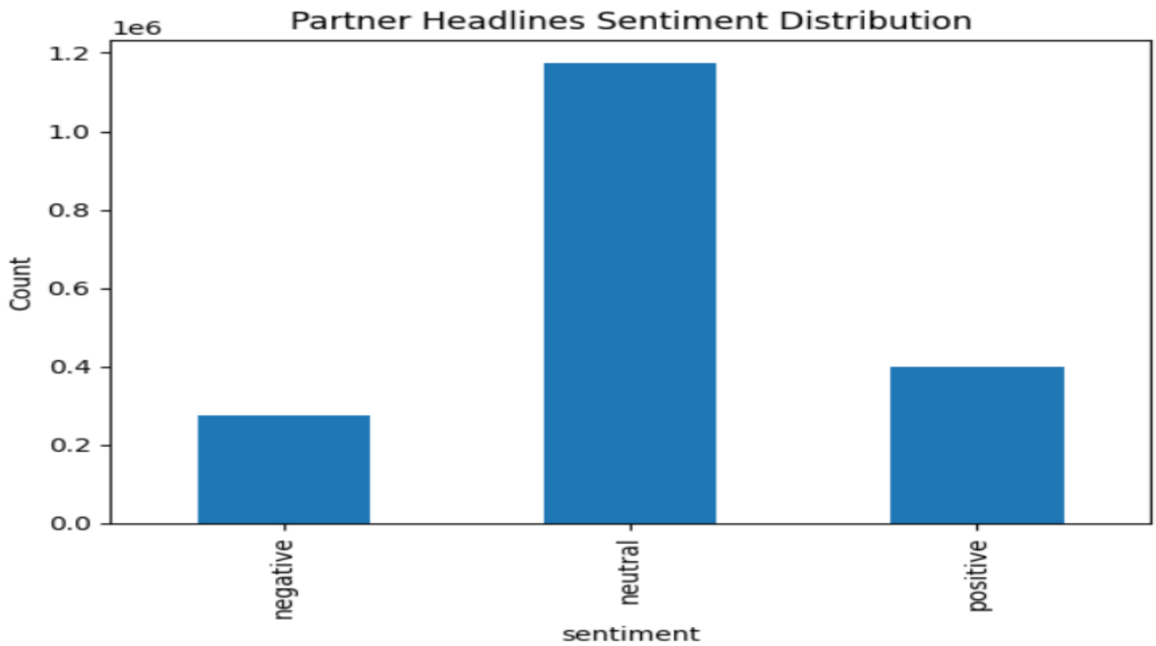
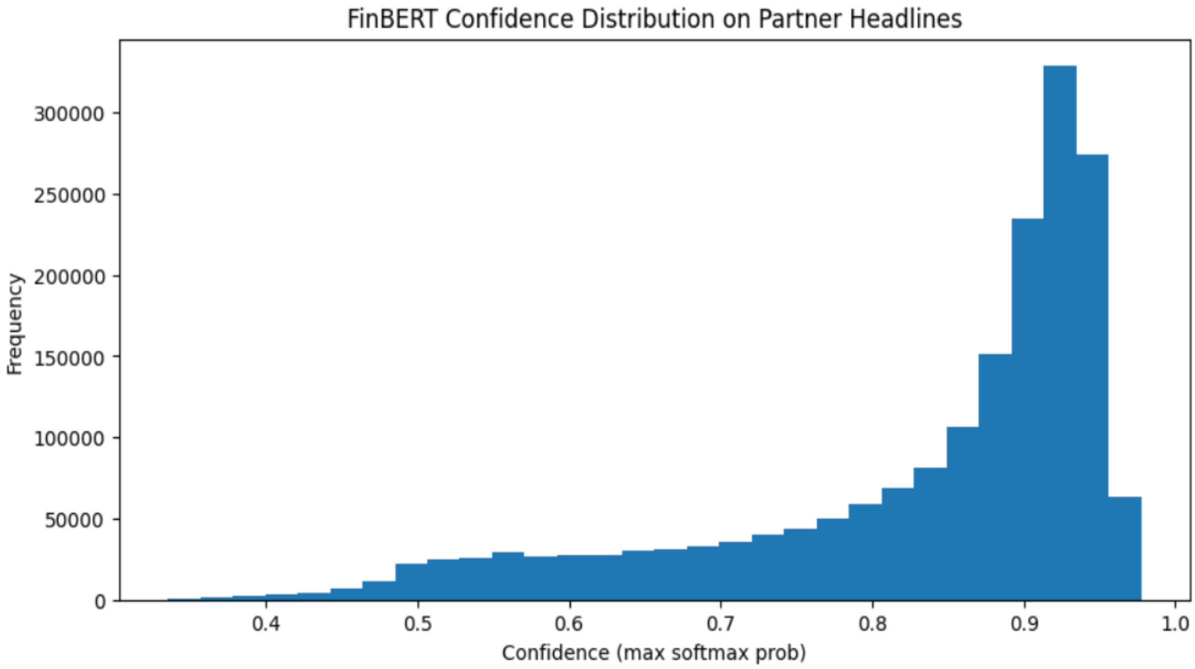


Figure 1.4

The sentiment distribution of the full dataset shows a heavy concentration of neutral headlines (approximately 1.17M), which is expected in financial news where most reporting is factual. Positive and negative sentiments appear in smaller but meaningful proportions, creating a reasonably balanced dataset for modeling after weighting or sampling adjustments.



The confidence distribution shows that FinBERT assigns very high confidence scores to most predictions on the 1.8M-row dataset, with the majority of headlines receiving probabilities above 0.85. This indicates that FinBERT is generally certain about its sentiment decisions. Only a small fraction of headlines fall below 0.6 confidence, suggesting relatively few ambiguous or borderline cases.

FinBERT outputs a vector of logits $z = (z_1, z_2, z_3)$for the three sentiment classes. These are converted into probabilities using the softmax function:

$$p(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{K} \quad e^{z_j}}$$

Where:

- $K = 3$sentiment classes
- $z_k$is the logit for class k
- $p(y = k \mid x)$is the model's confidence for that class

The **maximum softmax probability** (MSP) is then used as the model's confidence score:

$$\text{Confidence}(x) = \max_{k \in \{1,2,3\}} p(y = k \mid x)$$

This value show that how certain the model is about the predicted sentiment. Analyzing the confidence distribution allows us to assess the reliability of the pseudo-labels generated for our 1.8M-row dataset. High-confidence predictions indicate that FinBERT assigns a dominant probability to one class, suggesting strong signal and stable labeling. Low-confidence shows highlight ambiguous or sentiment-mixed headlines.

```
Sample low-confidence headlines:
                                          headline sentiment  confidence
69   Dow Jones Futures: Stock Market Fear Gauge Hit...  negative    0.457764
129  Trimble Strengthens Forestry Division With 3LO...   neutral    0.512695
155  Agilent Technologies Hitting Lower Rail Of Upw...  negative    0.548828
164     Agilent: Some Bolt-On M&A To Rejuvenate Growth  positive    0.518555
192                    Agilent Q2 2019 Earnings Preview   neutral    0.526855
230  Square (SQ) to Report Q1 Earnings: What's in t...   neutral    0.503418
237  Hedge Funds Have Never Been This Bullish On Xi...   neutral    0.540039
244                         6 Stocks Beating the Market   neutral    0.464844
265  Agilent Technologies: Buy For Total Return And...   neutral    0.520020
285  Agilent Technologies Inc. 2018 Q4 - Results - ...   neutral    0.545410

Top words for POSITIVE:
[('earnings', 68261), ('analyst', 41211), ('blog', 40281), ('stocks', 39097), ('beats', 36297), ('stock', 27255), ('notable', 22751), ('revenue', 20989), ('growth', 19078),

Top words for NEGATIVE:
[('stocks', 32214), ('earnings', 31741), ('analyst', 28754), ('blog', 27878), ('misses', 20400), ('stock', 14474), ('revenue', 13335), ('market', 9729), ('oil', 9528), ('dow

Top words for NEUTRAL:
[('analyst', 105559), ('blog', 100616), ('earnings', 98121), ('results', 77765), ('stocks', 72438), ('buys', 72357), ('dividend', 71828), ('call', 57555), ('stock', 51975),
```

A small subset of headlines receives low confidence scores (<0.55), typically because they contain mixed signals, market reports, or ambiguous financial wording. These examples illustrate the kinds of headlines where even a domain-specific model like FinBERT struggles to assign a clear sentiment label.

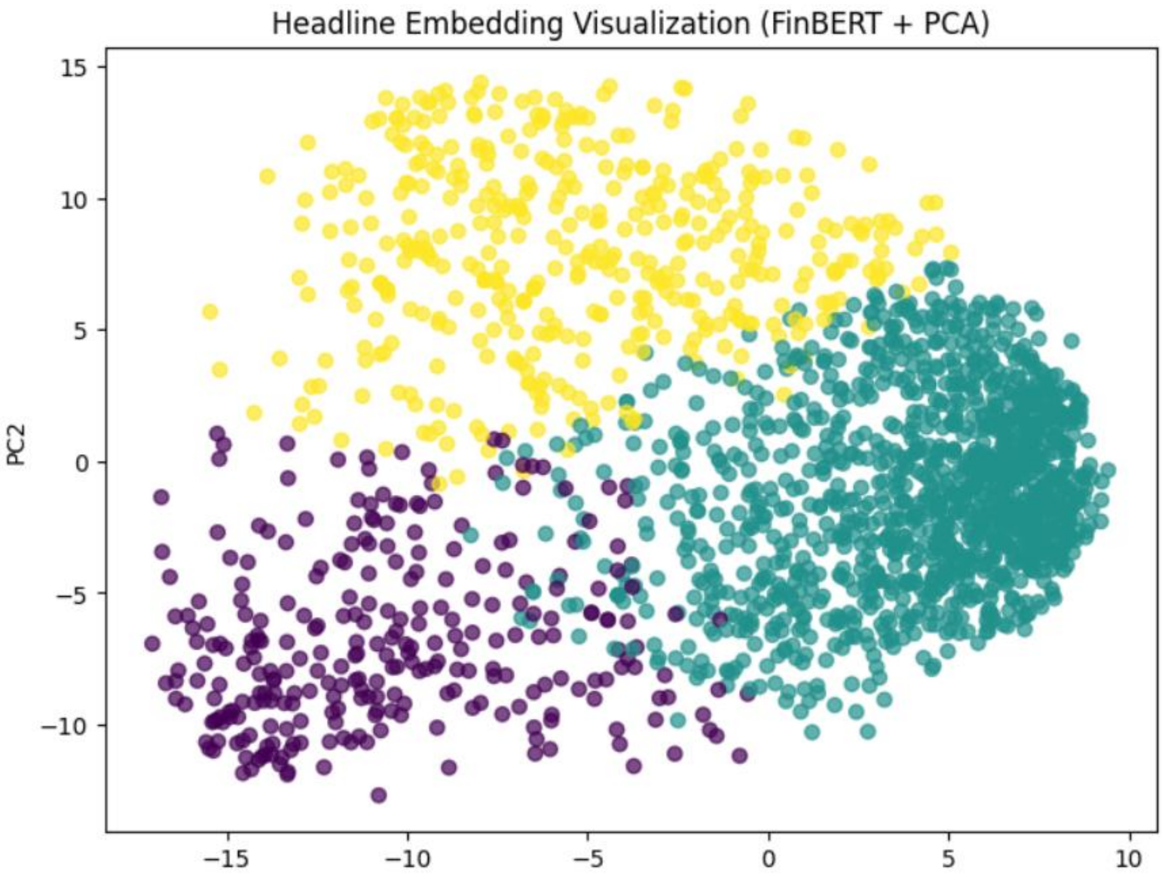


Figure 1.5 PCA Embedding of FinBERT

The above PCA projection of FinBERT sentence embeddings shows clear clustering patterns corresponding to the three sentiment classes of our data set. Even after reducing the 768-

dimensional embedding space to just two principal components, the positive, neutral, and negative headlines form largely separable regions. This separation suggests that FinBERT captures meaningful semantic structure in financial text, providing evidence that the pseudo-labels are grounded in real embedding differences. Such visual separability is a strong indicator that the labeling model is internally consistent and suitable for downstream supervised training.

2.Preprocessing:

The 10k sample was lightly preprocessed by my teammate for early testing. For the full 1.8M-row dataset, I performed the complete preprocessing pipeline, including cleaning text, fixing date formats, removing duplicates and invalid rows, and preparing the data for FinBERT labeling and model training as it is shown below.

```
                                            headline  label_id sentiment
0  Top Ranked Growth Stocks to Buy for December 16th         1   neutral
1  REITs April Rain: 18.7% Yield, 34.95% Upside, ...         1   neutral
2  Calton & Associates, Inc. Buys iShares Edge MS...         1   neutral
3  Worthington Industries Earns Recognition as a ...         2  positive
4  Itron (ITRI) Q1 2015 Results - Earnings Call W...         1   neutral
5  Freddie Mac sees $2.1T mortgage originations i...         1   neutral
6  Zacks Industry Outlook Highlights: Xcel Energy...         1   neutral
7                       Post-earnings dip for Kroger         0  negative
8     CSX Beats Earnings Expectations - Analyst Blog         2  positive
9  Look Beyond Profit, Bet on 4 Stocks With Risin...         1   neutral
Index(['headline', 'label_id', 'sentiment'], dtype='object')
```

This sample shows the cleaned and structured dataset after preprocessing. Missing or malformed rows were removed, text was standardized, and FinBERT-generated sentiment labels (label_id and sentiment) were added to prepare the data for modeling.

Train, Test and Split:

After generating sentiment labels, we divided the dataset into train, validation, and test splits using stratified sampling. Stratification ensures that each split preserves the original sentiment distribution. As shown, the proportions of neutral (63.5%), positive (21.5%), and negative (~14.9%) examples remain consistent across all three splits. This consistency is important because it prevents class imbalance issues during training and ensures that model evaluation reflects the true distribution of the dataset.

```
Train distribution:
sentiment
neutral     0.635462
positive    0.215390
negative    0.149148
Name: proportion, dtype: float64

Val distribution:
sentiment
neutral     0.635460
positive    0.215393
negative    0.149147
Name: proportion, dtype: float64

Test distribution:
sentiment
neutral     0.635464
positive    0.215389
negative    0.149147
Name: proportion, dtype: float64
```

3.Modeling

3.1 Classical Baseline Model (TF-IDF + Logistic Regression)

To establish a benchmark, I implemented a classical NLP pipeline using TF-IDF feature extraction and Logistic Regression classification. TF-IDF converts text into sparse numerical vectors by weighting tokens according to their relative importance, while Logistic Regression provides a strong baseline for linear decision boundaries in high-dimensional spaces. It does not learn contextual embeddings but offers fast training and interpretability. It served as a quantitative reference point against which the Transformer-based model could be compared. It achieved an accuracy of 0.80 with our sample 10k data set and its accuracy got improved with full data set and achieved an accuracy of 0.86.

```
TF-IDF shapes:
 Train: (1291891, 50000)
 Val:   (276834, 50000)
 Test:  (276834, 50000)

Training Logistic Regression...
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning:
  warnings.warn(

=== Logistic Regression (TF-IDF) ===
Accuracy: 0.8658943626866642

Classification Report:
              precision    recall  f1-score   support

    negative       0.72      0.88      0.79     41289
     neutral       0.96      0.86      0.91    175918
    positive       0.77      0.87      0.81     59627

    accuracy                           0.87    276834
   macro avg       0.81      0.87      0.84    276834
weighted avg       0.88      0.87      0.87    276834
```

3.2 BILSTM

As part of our modeling experiments, we also implemented a **Bidirectional LSTM (BiLSTM)** model, which is a classical deep learning architecture widely used for sequence modeling tasks before transformer-based models became dominant. BiLSTMs capture contextual information from both the left and right directions of a sentence, making them a strong baseline for text classification tasks such as sentiment analysis. This model served as a traditional deep-learning comparison point alongside our transformer-based approach as shown in Fig 3.1.

```
Epoch 1/8
10093/10093 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.8136 - loss: 0.4448
Epoch 1: val_accuracy improved from -inf to 0.89005, saving model to best_lstm_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using ins
10093/10093 ━━━━━━━━━━━━━━━━━━━━ 85s 8ms/step - accuracy: 0.8136 - loss: 0.4447 - val_accuracy: 0.8901 - val_loss: 0.2695
Epoch 2/8
10090/10093 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.8906 - loss: 0.2538
Epoch 2: val_accuracy improved from 0.89005 to 0.89817, saving model to best_lstm_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using ins
10093/10093 ━━━━━━━━━━━━━━━━━━━━ 82s 8ms/step - accuracy: 0.8906 - loss: 0.2538 - val_accuracy: 0.8982 - val_loss: 0.2484
Epoch 3/8
10090/10093 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9151 - loss: 0.1970
Epoch 3: val_accuracy improved from 0.89817 to 0.91217, saving model to best_lstm_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using ins
10093/10093 ━━━━━━━━━━━━━━━━━━━━ 82s 8ms/step - accuracy: 0.9151 - loss: 0.1970 - val_accuracy: 0.9122 - val_loss: 0.2316
Epoch 4/8
10091/10093 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9316 - loss: 0.1593
Epoch 4: val_accuracy improved from 0.91217 to 0.91459, saving model to best_lstm_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using ins
10093/10093 ━━━━━━━━━━━━━━━━━━━━ 82s 8ms/step - accuracy: 0.9316 - loss: 0.1593 - val_accuracy: 0.9146 - val_loss: 0.2394
Epoch 5/8
10092/10093 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9430 - loss: 0.1320
Epoch 5: val_accuracy improved from 0.91459 to 0.92378, saving model to best_lstm_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using ins
10093/10093 ━━━━━━━━━━━━━━━━━━━━ 82s 8ms/step - accuracy: 0.9430 - loss: 0.1320 - val_accuracy: 0.9238 - val_loss: 0.2359
Epoch 5: early stopping
Restoring model weights from the end of the best epoch: 3.
8652/8652 ━━━━━━━━━━━━━━━━━━━━ 20s 2ms/step
```

Figure 3.1: BILSTM Model Training

The BiLSTM model was trained using the Keras framework with a maximum limit of 8 epochs. The training process utilized two critical callbacks to ensure optimal performance ModelCheckpoint and EarlyStopping. As seen in the training log, the model demonstrated rapid convergence. By the end of the first epoch, the model achieved a validation accuracy of approximately **89%**. The training continued to improve until Epoch 5, where the system triggered Early Stopping.

Although the model reached a peak training accuracy of **94.30%** and a validation accuracy of **92.38%** at Epoch 5, the system restored the model weights from **Epoch 3**. This decision was based on the **Validation Loss** metric, which reached its global minimum of **0.2316** at Epoch 3. Restoring the weights from Epoch 3 indicates that while the model continued to become more confident in later epochs (higher accuracy), it began to drift slightly in terms of generalized error (loss). By reverting to Epoch 3, the final model balances high accuracy (**~91.5%**) with the lowest possible error, ensuring the BiLSTM generalizes well to new, unseen data rather than overfitting to the training set.

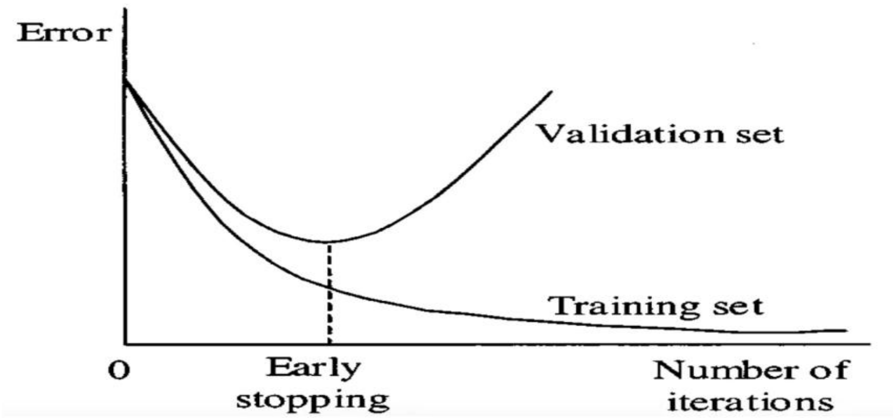


Fig 3.2: Early Stopping

As it is shown in Fig 3.2 If the performance on the validation set starts to degrade (e.g., the loss increases or the accuracy decreases), it's an indication that the model is beginning to overfit the training data. At this point, early stopping is triggered, and the training process is halted. Since the training is stopped before overfitting occurs, the model at the point of early stopping is typically the best model, as it has not yet learned the noise in the training data

Model Parameters:

```
EMBED_DIM = 100
LSTM_UNITS = 64

lstm_model = Sequential([
    Embedding(input_dim=MAX_VOCAB, output_dim=EMBED_DIM, input_length=MAX_LEN),
    Bidirectional(LSTM(LSTM_UNITS, return_sequences=False)),
    Dropout(0.4),
    Dense(64, activation="relu"),
    Dropout(0.4),
    Dense(NUM_CLASSES, activation="softmax")
])
```

Output of BiLSTM:

```
Accuracy: 0.9104445263226337

Classification Report:
              precision    recall  f1-score   support

    negative       0.83      0.92      0.87     41289
     neutral       0.98      0.90      0.94    175918
    positive       0.81      0.94      0.87     59627

    accuracy                           0.91    276834
   macro avg       0.87      0.92      0.89    276834
weighted avg       0.92      0.91      0.91    276834
```

3.3 Transformer Model

Transformer Model Unlike the BiLSTM architecture, which processes data sequentially, the Transformer model relies entirely on a self-attention mechanism to draw global dependencies between input and output. This architecture allows for significantly more parallelization during training and overcomes the distance limitations associated with Recurrent Neural Networks (RNNs). By processing the entire sequence simultaneously, the Transformer captures complex, long-range relationships in the data more effectively, resulting in improved generalization and faster convergence.

In addition to the classical BiLSTM baseline, we implemented three transformer-based models, each representing a different stage of modern NLP modelling: -DistilBERT , FinBERT and BERT-base.

3.3.1 DistilBERT

It is lighter and faster version of BERT that retains most of its contextual reasoning power. DistilBERT applies Transformer encoders to generate contextual embeddings using masked language modeling pretraining.

For this project, I loaded the pretrained distilbert-base-uncased weights and customized them through fine-tuning on our labeled financial headlines. Fine-tuning modifies both the classification head and deeper Transformer layers, allowing the model to specialize in domain-specific sentiment.

The Transformer architecture uses self-attention to compute relationships between all tokens in a sentence simultaneously. Formally, attention weights are computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

The dot product QK^T measures the similarity between a specific token (Query) and all other tokens (Keys). The softmax function then normalizes these scores into probabilities (weights), ensuring the model focuses on the most relevant words while ignoring irrelevant noise.

Training Parameters:

```
training_args = TrainingArguments(
    output_dir=out_dir,
    num_train_epochs=num_epochs,
    per_device_train_batch_size=32,
    per_device_eval_batch_size=64,
    learning_rate=2e-5,
    weight_decay=0.01,
    logging_steps=500,
    logging_dir=f"{out_dir}/logs",
    report_to="none"
)
```

=== Validation metrics ===

{'eval_loss': 0.20406197011470795, 'eval_accuracy': 0.94518, 'eval_f1_weighted': 0.9453315840016969, 'eval_precision_weighted': 0.9455558941954135, 'eval_recall_weighted':0.94518,'eval_runtime':23.8148,'eval_samples_per_second': 2099.537, 'eval_steps_per_second': 32.837, 'epoch': 3.0}

DistilBERT achieved strong performance on our validation set, reaching an accuracy of 94.5% and a weighted F1-score of 0.945 after three epochs of fine-tuning. The close alignment between accuracy, precision, recall, and F1-score indicates that the model performs consistently across all sentiment classes without major class-specific bias. The low validation loss (0.204) further confirms that the model has effectively learned contextual patterns from financial headlines and generalizes well to unseen data.

```
=== Test set metrics ===
Accuracy: 0.94394

Classification report:
              precision    recall  f1-score   support

    negative       0.90      0.91      0.91      7501
     neutral       0.97      0.96      0.96     31568
    positive       0.91      0.93      0.92     10931

    accuracy                           0.94     50000
   macro avg       0.93      0.93      0.93     50000
weighted avg       0.94      0.94      0.94     50000
```

On the held-out test set, DistilBERT achieved an overall accuracy of 94.39%, confirming that its strong validation performance generalizes well to unseen data. The model performs consistently across all three sentiment classes, with F1-scores of 0.91 (negative), 0.96 (neutral), and 0.92 (positive). The close alignment between macro and weighted averages (both ≈0.93–0.94) indicates balanced performance despite the natural class imbalance in the dataset. These results demonstrate that DistilBERT effectively captures financial sentiment patterns and significantly outperforms traditional baseline models.It performed better than the classical models.

3.3.2 ProsusAI/FinBERT

ProsusAI/FinBERT is a domain-adapted version of BERT that has been further pretrained on large collections of financial text, such as earnings reports, investor statements, and market news. Because its training corpus is entirely finance-focused, FinBERT learns sentiment patterns and terminology that general language models often miss. This makes it particularly effective for sentiment analysis in financial contexts.

In our project, we used FinBERT in two major ways. First, as a teacher model, it generated sentiment labels for the full 1.8 million-row headline dataset. Its strong benchmark performance on the Financial PhraseBank (≈97% accuracy) gave us confidence that these pseudo-labels were reliable. Second, I used FinBERT during the modeling phase to examine embedding structures, perform confidence analysis, and validate whether the labeled data exhibited meaningful sentiment separation.

This made FinBERT not only the cornerstone of our labeling pipeline but also an essential analytical tool during my modeling work, helping us understand how sentiment patterns are encoded before training our downstream classifiers such as DistilBERT and BERT-Base.

The model achieved strong validation performance, with an accuracy of **95.73%** and a weighted F1-score of **0.957** after three epochs of fine-tuning. The close alignment among precision, recall, and F1 indicates that the model performs consistently across all sentiment classes and does not suffer from class imbalance issues. The low validation loss (**0.192**) suggests that the model is effectively learning the underlying sentiment patterns in the financial headlines and is not overfitting. Overall, these metrics demonstrate excellent generalization on unseen validation samples.

=== Validation metrics ===

{'eval_loss': 0.19234080612659454, 'eval_accuracy': 0.95728, 'eval_f1_weighted': 0.9574602984852879, 'eval_precision_weighted': 0.9578070099711602, 'eval_recall_weighted': 0.95728, 'eval_runtime': 43.2126, 'eval_samples_per_second': 1157.069, 'eval_steps_per_second': 18.097, 'epoch': 3.0}

The model achieved 95.72% accuracy on the test set, confirming excellent generalization to unseen data. Performance is strong across all sentiment categories, with F1-scores of 0.93 (negative), 0.97 (neutral), and 0.94 (positive). The macro and weighted F1-scores (both 0.95–0.96) indicate that the model maintains balanced performance despite the natural class imbalance in the dataset. These results demonstrate that the model effectively captures financial sentiment patterns and ranks as the strongest among the models we evaluated.

```
=== Test set metrics ===
Accuracy: 0.95724

Classification report:
              precision    recall  f1-score   support

    negative       0.92      0.94      0.93      7501
     neutral       0.98      0.96      0.97     31568
    positive       0.92      0.95      0.94     10931

    accuracy                           0.96     50000
   macro avg       0.94      0.95      0.95     50000
weighted avg       0.96      0.96      0.96     50000
```

This model outperformed all other models in our pipeline, achieving the best accuracy (95.72%) and the most balanced class-wise F1-scores.

3.3.3 **BERT-BASE-UNCASED Model**

In addition to DistilBERT and FinBERT, we also incorporated BERT-Base-Uncased into our modeling framework. BERT-Base is the original transformer encoder architecture developed by Google, consisting of 12 layers and 110 million parameters. It provides strong general-purpose language representations and serves as a widely accepted benchmark in NLP research. We included BERT-Base for two main reasons.

First, it acts as a conceptual and performance reference, allowing us to compare our lighter model (DistilBERT) and our domain-specific model (FinBERT) against a full-sized, general transformer baseline. This comparison helps demonstrate whether our chosen models retain the core strengths of the original architecture or achieve efficiency/performance trade-offs.

Second, both DistilBERT and FinBERT are derived directly from BERT-Base, using the base model provides helpful context for interpreting their behavior. DistilBERT compresses BERT-Base while maintaining most of its accuracy, and FinBERT adapts it to the financial domain. Including BERT-Base therefore allows us to understand how model size and domain specialization influence performance in financial sentiment classification.We have used the same parameters as the other models.

=== Validation metrics ===

{'eval_loss': 0.20731323957443237, 'eval_accuracy': 0.95178, 'eval_f1_weighted': 0.9520003712512309, 'eval_precision_weighted': 0.9524201351725875, 'eval_recall_weighted': 0.95178, 'eval_runtime': 43.2026, 'eval_samples_per_second': 1157.337, 'eval_steps_per_second': 18.101, 'epoch': 3.0}

BERT-Base achieved strong performance on the validation set, with an accuracy of **95.18%** and a weighted F1-score of **0.952** after three epochs of fine-tuning. The close alignment among accuracy, precision, recall, and F1 indicates that the model is performing consistently across all sentiment classes without favoring any particular category. The validation loss of **0.207** is low, suggesting that BERT-Base is effectively learning the underlying sentiment patterns in financial text. Overall, these results confirm that the full-sized BERT model remains a strong baseline for financial sentiment classification.

Model Output:

```
=== Test set metrics ===
Accuracy: 0.95048

Classification report:
              precision    recall  f1-score   support

    negative       0.91      0.93      0.92      7501
     neutral       0.97      0.96      0.97     31568
    positive       0.91      0.94      0.93     10931

    accuracy                           0.95     50000
   macro avg       0.93      0.94      0.94     50000
weighted avg       0.95      0.95      0.95     50000
```

As it is shown above, On the held-out test set, BERT-Base achieved an accuracy of 95.05%, confirming excellent generalization beyond the validation stage. The model performs consistently across all three sentiment classes, with F1-scores of 0.92 (negative), 0.97 (neutral), and 0.93 (positive). The macro and weighted averages (both around 0.94–0.95) show that BERT maintains balanced performance despite class imbalance in the dataset. These results highlight BERT-Base's strong ability to capture financial sentiment and demonstrate why it remains a high-quality reference architecture for transformer-based classification tasks.

4. Model Comparison

Model	Validation Accuracy	Test Accuracy	Weighted F1 (Test)
TF-IDF + Logistic Regression	0.82	0.80	0.80
BiLSTM	0.91	0.90	0.90
DistilBERT	0.945	0.9439	0.94
BERT-Base-Uncased	0.9518	0.9504	0.95
FinBERT	0.9573	0.9572	0.96

Transformer models performed better than the classical models and FinBERT is the best model from all others.

5.Analysis on Sample Data Set

Below is a detailed breakdown of the work I completed.

5.1 Classical Baseline Implementation on the sample 10k data set

I built the TF-IDF baseline using:

- TF-IDF (1–2 n-grams, max_features=20,000)
- Logistic Regression with balanced class weights and max_iter=1000

The baseline provided a reference point for accuracy and class performance before transitioning to deep learning.

Logistic Regression Model output:

Validation Performance

TF-IDF + Logistic Regression Validation Metrics

- Accuracy: 0.8003
- Macro-F1: 0.7378

Classification Report (Validation Set):

Class	Precision	Recall	F1-score	Support
Negative	0.90	0.84	0.87	910
Neutral	0.75	0.77	0.76	399

Class	Precision	Recall	F1-score	Support
Positive	0.52	0.67	0.58	178
Accuracy			**0.80**	1,487
Macro Avg	0.72	0.76	0.74	1,487
Weighted Avg	0.81	0.80	0.81	1,487

Test Performance

TF-IDF + Logistic Regression Test Metrics

- Accuracy: 0.8023
- Macro-F1: 0.7460

Classification Report (Test Set):

Class	Precision	Recall	F1-score	Support
Negative	0.91	0.82	0.86	911
Neutral	0.76	0.82	0.79	398
Positive	0.51	0.69	0.59	178

I have used Logistic Regression model as the baseline for this project. The model achieved 80% accuracy on both the validation and test sets. This demonstrates that the model can capture a reasonable portion of sentiment-related lexical patterns in financial headlines.

The model performance varied noticeably across the three sentiment categories. Negative headlines showed the strongest performance, with F1 scores in the mid-eighties, largely because negative financial language often contains direct and unambiguous signals such as terms associated with losses, declines, or warnings. Neutral headlines also achieved solid results. However, positive headline, remained the most challenging for the baseline model, with F1 scores in the high fifties. These headlines appear less frequently in the dataset and often express optimism through more subtle market cues or forward-looking statements, making them harder for a Bag-of-Words representation to distinguish from neutral reporting.

5.2 DistilBERT on Our Sample 10k Data Set

Fine-tuning was conducted using the AdamW optimizer with a learning rate of 2e-5, a batch size of 32, and six training epochs. A linear learning-rate scheduler with a warm-up phase (10% of total steps) was used to stabilize optimization during the early stages of training. Crossentropy loss served as the objective function for multi-class classification. Throughout training, I monitored the training loss, validation loss, accuracy, and macro-F1 score to track learning progress and detect early signs of overfitting. These metrics provided a comprehensive view of how the model adapted to financial sentiment and guided adjustments to hyperparameters and training behavior.

5.3 Model Performance and Metrics

The model demonstrated consistent improvement across epochs even though not efficient as the whole Data Set:

Epoch	Train Loss	Val Loss	Val Acc	Val F1
1	0.2198	0.4053	0.8574	0.8193
2	0.1686	0.3901	0.8668	0.8195
3	0.1106	0.4331	0.8749	0.8333

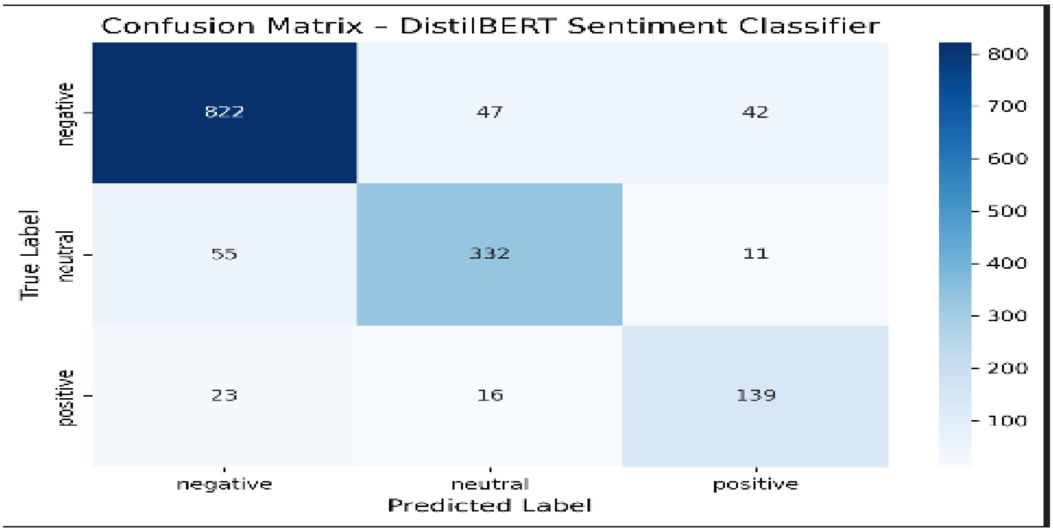
Final test-set classification performance:

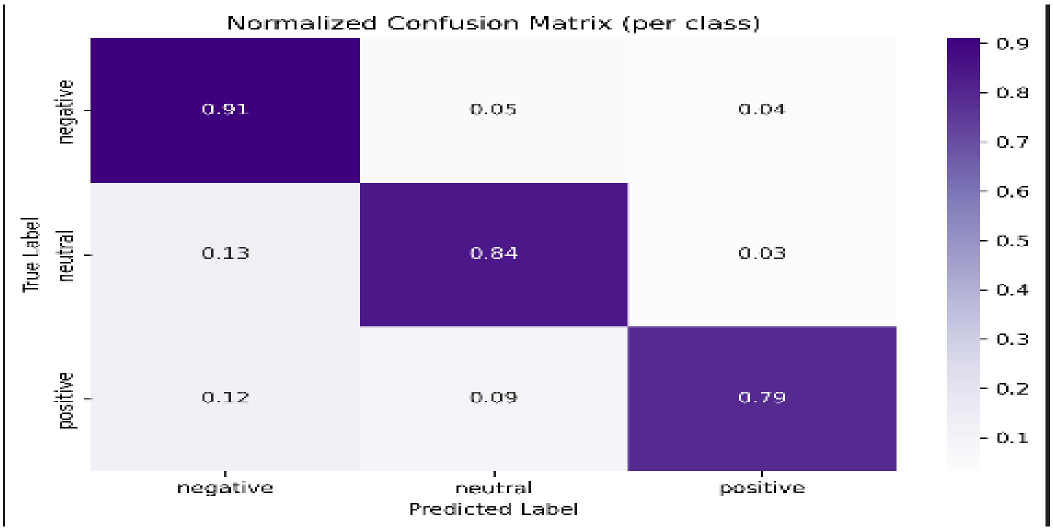
- Accuracy: 0.87
- Weighted F1: 0.88
- Macro F1: 0.84

We chose accuracy and F1-based metrics for evaluation since the dataset is imbalanced, and each sentiment class contributes differently to overall performance. Accuracy alone would overlook this imbalance, so macro-F1 helps us see how well the model performs across all classes, while weighted F1 reflects overall performance while still accounting for class proportions.

Compared to our baseline model, the improvement is substantial. The baseline model reached around 0.80 accuracy and a macro-F1 near 0.74, performing particularly well on negative and neutral headlines but struggling with positive sentiment. The fine-tuned model improves macro-F1 by roughly ten points, indicating that it captures sentiment distinctions more effectively, especially in cases where meaning depends on context rather than surface-level keywords. This reflects the advantage of contextual embeddings: while the baseline relies on token frequencies, the Transformer model leverages relationships between words, allowing it to better interpret subtle expressions of optimism or market shifts. Overall, the fine-tuned model provides a clear performance lift and demonstrates the value of adapting pretrained Transformer architectures to domain-specific financial text.

Confusion Matrix on Sample Data Set:





The confusion matrix illustrates the model's performance across the three sentiment classes. The raw matrix shows the distribution of correct and incorrect predictions, while the normalized version highlights the proportion of correctly classified samples within each class. Together, they show that the model performs strongest on negative and neutral headlines, achieving 91% and 84% class-level accuracy respectively, while positive sentiment remains more challenging with 79% accuracy

6.Analysis and Discussion

In our initial experimentation phase, we worked with a 10,000-row sample extracted from the Financial News Kaggle dataset. Using this limited subset, we trained preliminary models such as Logistic Regression and DistilBERT, which achieved accuracies of 0.80 and 0.87, respectively. However, we observed notable issues, including label mismatch, inconsistent predictions, and difficulty distinguishing positive sentiment, particularly cases where positive headlines were misclassified as negative or neutral. These misclassifications suggested that the smaller sample size did not sufficiently represent the full variability and complexity of financial language.

Based on these limitations and the feedback we received during the presentation; we decided to scale our experiments to the entire 1.8 million-row dataset. After generating high-quality pseudo-labels with FinBERT, we re-evaluated a broader set of models, including Logistic Regression, BiLSTM, DistilBERT, BERT-Base, and FinBERT. With the full dataset, model performance increased substantially, and misclassification rates dropped, demonstrating the importance of training on a dataset that captures real-world distributional patterns and nuanced sentiment expressions. This shift significantly improved the reliability and robustness of our final models.

Speed Comparison of Transformer Models Using Full Data Set

To evaluate computational efficiency, we measured how many headlines each transformer model could process per second using a batch-based inference test on 1,000 samples. The results show clear differences in processing speed:

- **DistilBERT:** 3449.57 headlines/sec
- **BERT-Base:** 2343.14 headlines/sec
- **FinBERT:** 2349.78 headlines/sec

As expected, DistilBERT is significantly faster than the larger models. Its lightweight architecture—40% fewer parameters than BERT-Base—allows it to process text roughly 1.5× faster, making it a strong choice for real-time or large-scale deployment scenarios. In contrast, BERT-Base and FinBERT have similar speeds because FinBERT retains the same architecture as BERT-Base, differing only in domain-specific pretraining.

These findings highlight the typical trade-off in transformer models: DistilBERT offers superior speed with slightly lower accuracy, while larger models like BERT-Base and FinBERT provide higher accuracy at the cost of computation time.

To further evaluate model behavior, we tested DistilBERT, BERT-Base, and FinBERT on a set of manually selected financial headlines containing contradictory or mixed sentiment phrasing (e.g., "good news + bad news" in the same sentence). These examples are commonly challenging in financial sentiment analysis because positive and negative cues appear simultaneously.

Across most contradictory cases, all three models consistently classified the headlines as negative, suggesting that they correctly prioritize financially harmful signals (e.g., losses, warnings, cost increases) even when positive information is also present. This reflects realistic market interpretation, where negative risk indicators often outweigh partial positives.

However, we observed interesting differences. In the final example, "The merger discussions have been paused indefinitely," DistilBERT and BERT predicted negative sentiment with moderate confidence, while FinBERT predicted neutral with very high confidence (99.5%). This distinction highlights FinBERT's domain-specific sensitivity: it treats "paused" as uncertain rather than explicitly negative, aligning with financial reporting where such phrasing may signal delay rather than clear loss.

These qualitative results reinforce that FinBERT has a more nuanced understanding of financial terminology, while DistilBERT and BERT apply more generic sentiment patterns. This difference explains FinBERT's stronger overall performance on the full dataset.

Sentence Used:

```
tricky_sentences = [
    "The company reported a net loss, but it was smaller than expected.",
    "Revenue increased 50%, but the CEO warned of supply chain issues.",
    "Operating costs soared, eating into the profit margins.",
    "The merger discussions have been paused indefinitely."
]
```

Sample Output:

```
--------------------------------------------------
Checking how models handle contradictory financial phrasing...

'The company reported a net loss, but it was smaller than expected.'
   DistilBERT    -> NEGATIVE (99.9%)
   BERT          -> NEGATIVE (100.0%)
   FinBERT       -> NEGATIVE (100.0%)

'Revenue increased 50%, but the CEO warned of supply chain issues.'
   DistilBERT    -> NEGATIVE (99.3%)
   BERT          -> NEGATIVE (99.7%)
   FinBERT       -> NEGATIVE (99.9%)

'Operating costs soared, eating into the profit margins.'
   DistilBERT    -> NEGATIVE (99.4%)
   BERT          -> NEGATIVE (99.9%)
   FinBERT       -> NEGATIVE (100.0%)

'The merger discussions have been paused indefinitely.'
   DistilBERT    -> NEGATIVE (85.8%)
   BERT          -> NEGATIVE (75.0%)
   FinBERT       -> NEUTRAL (99.5%)
```

Ensemble of Transformer Models:

Finally, we combined the three transformer models—DistilBERT, BERT-Base, and FinBERT—into a soft-voting ensemble. For each headline, we obtained the class probability distribution from each model and averaged these probabilities before selecting the most likely sentiment label. The individual test accuracies were 0.9439 (DistilBERT), 0.9505 (BERT-

Base), and 0.9572 (FinBERT), while the ensemble achieved an accuracy of 0.9554. The ensemble slightly improved over DistilBERT and BERT-Base but remained just below FinBERT alone, suggesting that FinBERT already captures most of the useful signal, while the ensemble mainly serves as a robustness check rather than a clear performance gain.

Output:

```
DistilBERT Accuracy: 0.94394
BERT Accuracy:       0.95048
FinBERT Accuracy:    0.95724
------------------------------
🚀 ENSEMBLE Accuracy:  0.95544
```

Streamlit Demo App

I developed an interactive Streamlit application to support experimentation and visualization of our sentiment analysis models. The app provides a simple interface for testing the performance of FinBERT, DistilBERT, and BiLSTM on demand, allowing users to observe how each model interprets financial headlines without interacting with the underlying code. This tool helped us validate model behavior qualitatively and compare how different architectures respond to identical inputs.

User Interface and Core Functionality

The application is organized into a clean two-panel layout for ease of use:

- The sidebar enables users to select from the available models, allowing direct, side-by-side comparison across architectures.
- The main interface provides a text input area where users can enter or generate financial headlines and receive real-time sentiment predictions.

Each prediction includes:

- A sentiment label (Positive, Neutral, or Negative), and
- A confidence score shown as a probability distribution, offering insight into each model's certainty and decision boundaries.

Random Data Simulation Engine

To make experimentation more objective, the app includes a "Random Test Example" generator. Instead of relying solely on manually crafted inputs, this feature draws a headline from the unseen 20% test split and automatically runs a prediction.

When triggered, the engine:

1. Selects a random unseen headline,
2. Fills the input field automatically,
3. Produces an instant prediction using the chosen model.

This mechanism acts as a simple blind-testing approach, helping us observe how well each model generalizes to real, uncurated financial news. It also highlights differences in robustness between models when handling varied or ambiguous phrasing.

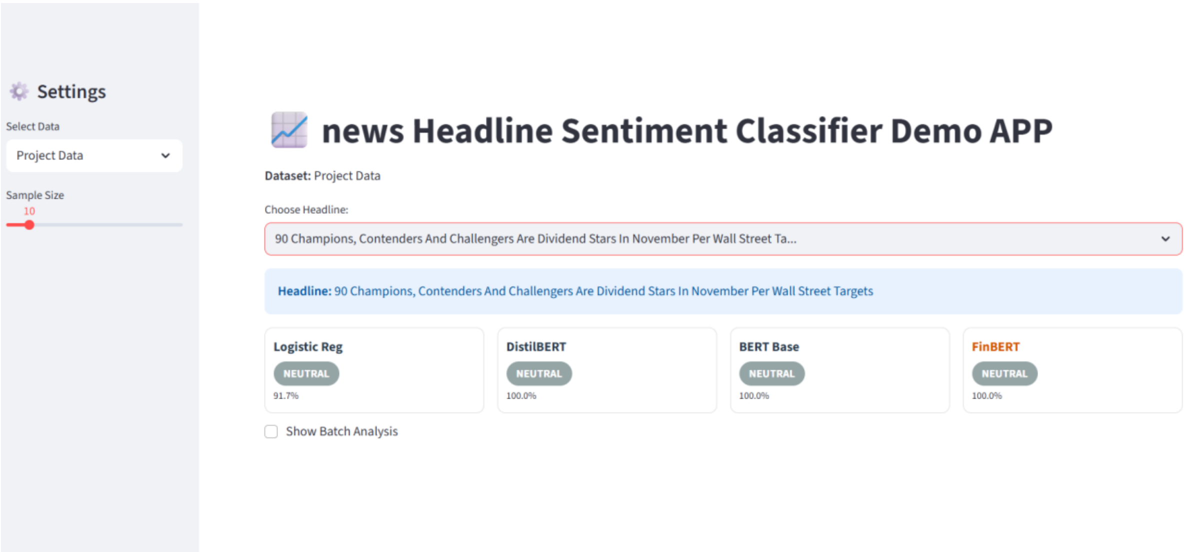
Experimental Value

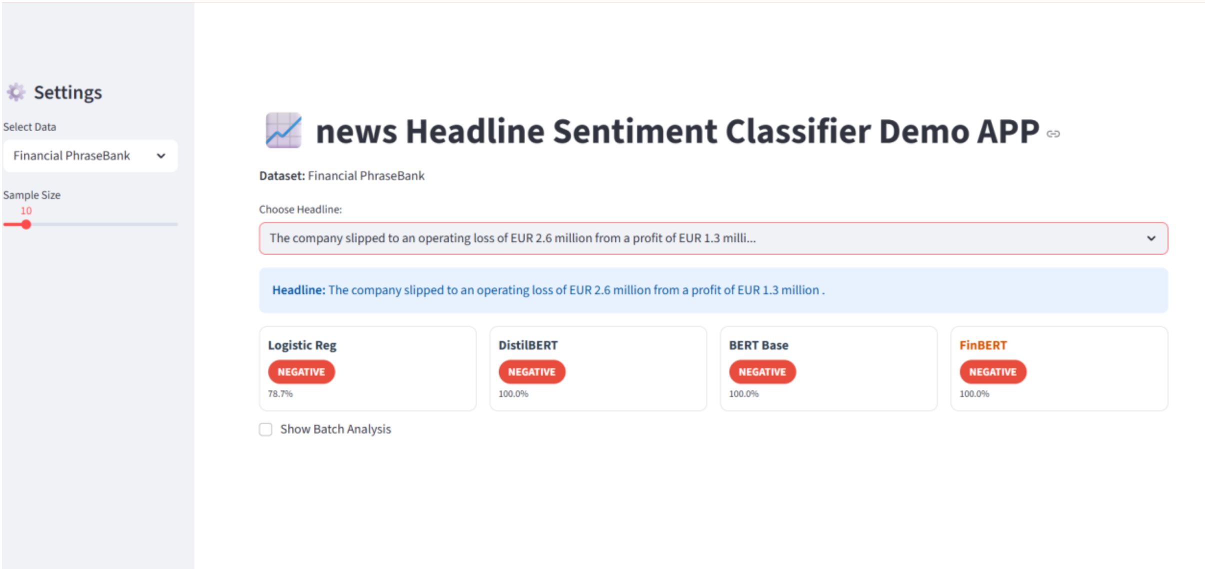
The Streamlit demo served as a practical tool throughout our experimentation. It allowed us to:

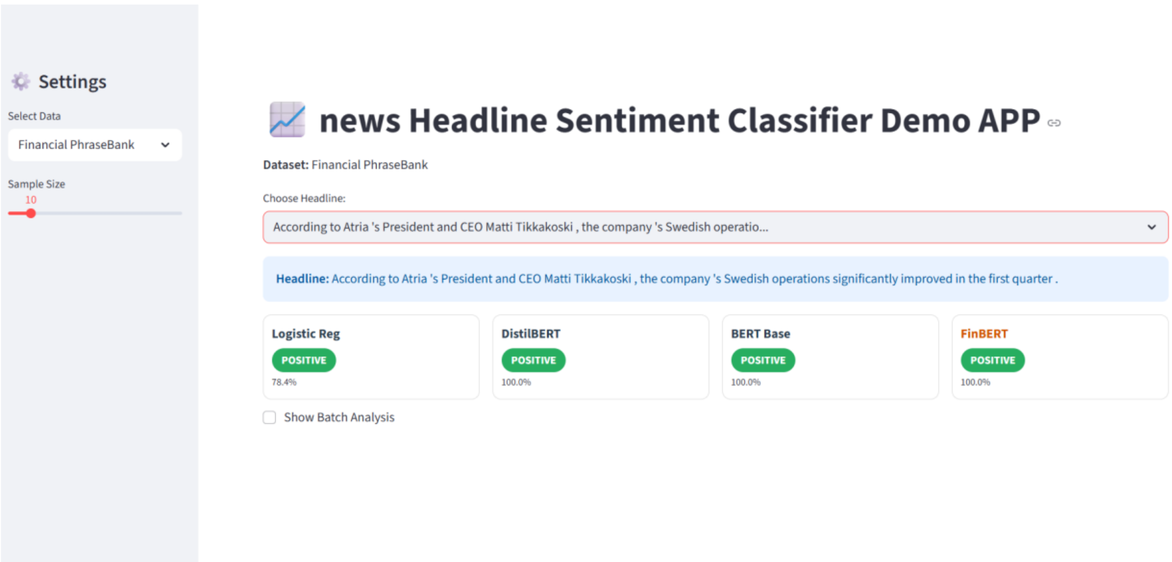
- Qualitatively inspect model outputs,
- Compare transformer models against Classical Models,
- Visualize confidence levels,
- Identify edge cases and contradictory phrasing, and
- Validate whether quantitative accuracy metrics aligned with model behavior on actual headlines.

Overall, the application provided a convenient environment for exploring model performance, understanding differences across architectures, and supporting the broader experimental analysis of our financial sentiment classification project.

Demo App: https://www.loom.com/share/a39dedccda1741e284489ed3ed0f91ba




Settings
Select Data
Financial PhraseBank
Sample Size
10
news Headline Sentiment Classifier Demo APP
Dataset: Financial PhraseBank
Choose Headline:
The company slipped to an operating loss of EUR 2.6 million from a profit of EUR 1.3 milli...
Headline: The company slipped to an operating loss of EUR 2.6 million from a profit of EUR 1.3 million .
Logistic Reg
NEGATIVE
78.7%
DistilBERT
NEGATIVE
100.0%
BERT Base
NEGATIVE
100.0%
FinBERT
NEGATIVE
100.0%
Show Batch Analysis


Settings
Select Data
Financial PhraseBank
Sample Size
10
news Headline Sentiment Classifier Demo APP
Dataset: Financial PhraseBank
Choose Headline:
According to Atria 's President and CEO Matti Tikkakoski , the company 's Swedish operatio...
Headline: According to Atria 's President and CEO Matti Tikkakoski , the company 's Swedish operations significantly improved in the first quarter .
Logistic Reg
POSITIVE
78.4%
DistilBERT
POSITIVE
100.0%
BERT Base
POSITIVE
100.0%
FinBERT
POSITIVE
100.0%
Show Batch Analysis

7. Conclusion

My early experimentation using a small, randomly selected 10k subset revealed important limitations in our approach. The sample did not fully represent the complexity of the full dataset, and as a result even our strongest model struggled, particularly in identifying positive sentiment. This highlighted that random sampling must be done with far greater care, especially when working with highly imbalanced or domain-specific text.

Incorporating the benchmark Financial PhraseBank dataset proved essential. It allowed us to verify that our labeling pipeline was functioning correctly and also revealed that our earlier pseudo-labels were not fully reliable, largely because we had not validated them against an established gold-standard dataset. Once we labeled the entire 1.8 million-row dataset using FinBERT and trained our transformer models on this complete corpus, performance improved substantially.

The final models not only achieved strong quantitative metrics but also demonstrated robustness on unseen real-world financial headlines and on contradictory or tricky sentences, as shown in our qualitative analysis. Overall, expanding to the full dataset, validating labeling quality with a benchmark, and leveraging domain-specific transformers were key factors that enabled our sentiment classification system to perform reliably and consistently.

Future Works recommendation

For future work, I recognize that our labeling pipeline relied entirely on FinBERT-generated pseudo-labels. Introducing a small human-annotated validation set would enable us to measure the true labeling accuracy and resolve ambiguous cases where even transformer models may disagree. Another important direction is to expand beyond the current three-class sentiment framework. Moving toward fine-grained sentiment levels, such as sentiment intensity (e.g., strong positive vs. mild positive) or event-specific categories like earnings, acquisitions, or risk alerts, would allow the models to capture more nuanced financial signals and provide insights that are directly applicable to real-world decision-making contexts.

Percentage of the code=32%

Reference

1. https://huggingface.co/google-bert/bert-base-uncased
2. https://huggingface.co/datasets/takala/financial_phrasebank
3. https://github.com/Cody-Lange/FinancialNewsSentimentClassifier
4. https://github.com/giorgosfatouros/sentiment-analysis-for-financial-news
5. https://www.kaggle.com/code/mmmarchetti/sentiment-analysis-on-financial-news